

Diffie-Hellman Key Exchange

1. Aim

To implement the Diffie-Hellman key exchange algorithm and to understand how two parties can establish a shared secret key over an insecure communication channel without ever transmitting the key itself.

2. Algorithm

1. Start the program.
2. Read a prime number p and a primitive root g entered by the user.
3. Read Alice's private key a and Bob's private key b entered by the user.
4. Compute Alice's public key using $A = g^a \bmod p$.
5. Compute Bob's public key using $B = g^b \bmod p$.
6. Exchange the public keys A and B between Alice and Bob over the channel.
7. Compute the shared secret key on Alice's side as $B^a \bmod p$.
8. Compute the shared secret key on Bob's side as $A^b \bmod p$.
9. Display both public keys and both shared secret keys, verify that they are equal, and stop the program.

3. Program

```

#include <stdio.h>

long long power(long long a, long long b, long long mod) {
    long long result = 1;
    while (b > 0) {
        if (b % 2 == 1)
            result = (result * a) % mod;
        a = (a * a) % mod;
        b /= 2;
    }
    return result;
}

int main() {
    long long p, g, a, b, A, B;

    printf("Enter Prime Number: ");
    scanf("%lld", &p);
    printf("Enter Primitive Root: ");
    scanf("%lld", &g);
    printf("Enter Alice Private Key: ");
    scanf("%lld", &a);
    printf("Enter Bob Private Key: ");
    scanf("%lld", &b);

    A = power(g, a, p);
    B = power(g, b, p);

    printf("Alice Public Key: %lld\n", A);
    printf("Bob Public Key: %lld\n", B);
    printf("Shared Secret by Alice: %lld\n", power(B, a, p));
    printf("Shared Secret by Bob: %lld\n", power(A, b, p));

    return 0;
}

```

4. Output Screenshot

The screenshot shows a code editor with a dark theme. On the left, there's a sidebar with icons for file explorer, search, and other IDE features. The main area displays the C code from the previous block. On the right, there's an 'Output' panel with a 'Clear' button. The output text shows the program's execution: it prompts for and reads the prime number (29), primitive root (2), Alice's private key (5), and Bob's private key (9). It then calculates and displays Alice's public key (3), Bob's public key (19), and the shared secret (21) for both parties. At the bottom of the output, it says '=== Code Execution Successful ==='. Below the code editor, a URL is visible: https://eb2.3liff.com/pass?tl_clickthrough=true&redir=https%3A%2F%2Ftrb-...

5. Sample Input / Output

```
Enter Prime Number: 23
Enter Primitive Root: 5
Enter Alice Private Key: 6
Enter Bob Private Key: 15
Alice Public Key: 8
Bob Public Key: 19
Shared Secret by Alice: 2
Shared Secret by Bob: 2
```

6. Result

The Diffie-Hellman key exchange algorithm was implemented and executed successfully, and both parties were shown to arrive at the same shared secret key without transmitting it directly.